Moderated Latent Variables Mediation Using Correct and Misspecified Models

Felipe Fontana Vieira and Yves Rosseel

Department of Data Analysis, Faculty of Psychology and Educational Sciences, Ghent University

Author Note

Felipe Fontana Vieira https://orcid.org/0009-0006-0949-6569

Yves Rosseel https://orcid.org/0000-0002-4129-4477

This study was preregistered. The preregistration, data, code, and materials for this manuscript can be found at https://doi.org/10.5281/zenodo.20703258. This project was funded by the Special Research Fund (BOF) of Ghent University (BOF/24J/2023/144). Author roles were classified using the Contributor Role Taxonomy (CRediT; https://credit.niso.org/) as follows: Felipe Fontana Vieira: Conceptualization, Methodology, Software, Investigation, Data curation, Formal analysis, Visualization, Writing - original draft, Writing - review & editing; Yves Rosseel: Conceptualization, Methodology, Software, Supervision, Writing - review & editing, Funding acquisition

Correspondence concerning this article should be addressed to Felipe Fontana Vieira, Department of Data Analysis, Faculty of Psychology and Educational Sciences, Ghent University, Henri Dunantlaan 2, Ghent 9000, Belgium, Email: felipe.fontanavieira@ugent.be

Abstract

TODO: Abstract.

Translational Abstract

TODO: Translational abstract.

Keywords: moderated mediation, structural equation modeling, latent interaction, measurement error, model misspecification

Moderated Latent Variables Mediation Using Correct and Misspecified Models

Mediation analysis is a well-known procedure in the behavioral and social sciences (Charlton et al., 2021; Hayes, 2018; Pieters, 2017; Preacher et al., 2007). In its basic and standard form, the effect of an independent variable on an outcome is decomposed into a direct effect and an indirect effect transmitted through a mediator, with all variables observed. This basic single-mediator model has since been extended, for example by allowing the indirect effect to depend on a moderator and by adjusting for covariates (Edwards & Lambert, 2007; Hayes, 2015; Montoya, 2024; Preacher et al., 2007).

A more fundamental concern arises from measurement. Many constructs in psychology are not directly observed but are measured indirectly through multiple items, that is, they are latent variables (Bollen & Hoyle, 2012; Ledgerwood & Shrout, 2011). When such constructs are reduced to summary scores (e.g., a sum or mean of the items) and analyzed as though they were observed, measurement error is ignored. This error can bias the indirect effect and inflate its uncertainty (Cole & Preacher, 2014; Hoyle & Kenny, 1999), and it is especially problematic for the mediator, which acts at once as a predictor and an outcome (Gonzalez & MacKinnon, 2021). To address this, SEM with latent variables has been proposed (**cite?**), especially in the context of mediation (**cite?**). It counteracts the attenuation by estimating structural relationships between common factors, from which indicator-specific unique variance has been partitioned (Bollen, 1989), provided the measurement model is correctly specified (**cite?**). The problem is compounded under moderation, where the product term is less reliable than either of its constituents (Bohrnstedt & Marwell, 1978; Busemeyer & Jones, 1983).

Yet, combining latent measurement with moderation yields a latent-variable moderated mediation model (Hayes, 2015), which is not straightforward in terms of estimation. A latent interaction involves the product of two latent variables, which is unobserved and nonnormal even when its constituents are normal (Klein & Moosbrugger, 2000). A number of methods have been proposed to estimate latent interactions (Algina & Moulder, 2001; Arminger & Muthén, 1998; Asparouhov & Muthén, 2021; Boker et al., 2023; Bollen, 1995; Cox & Kelcey, 2021; Holst &

Budtz-Jørgensen, 2020; Jöreskog & Yan, 1996; Kenny & Judd, 1984; Klein & Moosbrugger, 2000; Klein & Muthén, 2007; Marsh et al., 2004; Mooijaart & Bentler, 2010; Mooijaart & Satorra, 2012; Ng & Chan, 2020; Rosseel & Loh, 2024; Wall & Amemiya, 2000, 2001, 2003). Most adopt a system-wide (joint) estimation strategy, in which the measurement and structural parameters are estimated simultaneously, and include the most known methods: latent moderated structural equations (LMS) and product-indicator (PI). Opposed to these are structural-after-measurement (SAM) approaches (Cox & Kelcey, 2021; Holst & Budtz-Jørgensen, 2020; Ng & Chan, 2020; Rosseel & Loh, 2024), in which the measurement model is estimated first, from the indicators alone, and the structural model is estimated afterward with the measurement parameters held fixed. Until recently, these abovementioned approaches were rarely applied in practice, held back by limited software, slow estimation, and error-prone implementations. This has changed with recent open-source implementations (Holst & Budtz-Jørgensen, 2020; Rosseel et al., 2025; Slupphaug et al., 2024) and non-commercial software (Keller, 2025).

In this context, a further consideration concerns interpretation. Mediation is a hypothesis about causal effects, in that the indirect effect is intended as the effect of the predictor on the outcome that is transmitted through the mediator. A causal interpretation requires identification assumptions that cannot be verified from the data alone (Imai et al., 2010; Rohrer et al., 2022; VanderWeele, 2015). With latent variables, further requirements apply (Sterner et al., 2024; VanderWeele & Vansteelandt, 2022). We focus on one that is central to interpretation: the estimated latent variables must validly represent the intended constructs (Gonzalez & MacKinnon, 2021). When they do not, the model is subject to interpretational confounding (Burt, 1976), a discrepancy between the empirical meaning assigned to a latent variable through its estimated measurement parameters and the meaning intended by the researcher. In one-step SEM, where the measurement and structural parameters are estimated simultaneously, a misspecification in the structural model can propagate to the measurement model and alter the estimated latent variables. In moderated mediation, such misspecifications are plausible (e.g., an omitted moderator effect on the outcome, an omitted moderated direct effect, a cross-loading, or a residual mediator-outcome covariance), and each can distort the constructs whose conditional indirect effect is of interest. SAM avoids this problem. Because the structural model cannot influence the measurement model, structural misspecification does not affect the estimated latent variables, and interpretational confounding is avoided, though this does not imply that the structure is correct or fixed. Local SAM (LSAM) is one variant of this approach, and it has recently been extended to interaction and quadratic effects among latent variables (Rosseel et al., 2025), which makes it applicable to latent-variable moderated mediation.

To illustrate, for simplicity, we take a mediation model with three latent variables, X, M, and Y, each with three indicators, and obtain a population covariance matrix from a correctly specified version. We then alter it: the covariances between the X and M indicators are made unequal across the X indicators (3.0, 1.5, and 1.5, with mean held at the implied value of 2.0), so that only their pattern, not the overall X-M strength, changes. We fit the model by one-step SEM and by SAM (N = 200).

Table 1

Illustration for Interpretational Confounding.

Parameter	True	One-step SEM	SAM
Loading of x_1 on X	1.00	1.00	1.00
Loading of x_2 on X	1.00	0.85	1.00
Loading of x_3 on X	1.00	0.85	1.00
$M \sim X$	0.50	0.53	0.50
$Y \sim M$	0.50	0.53	0.50
$Y \sim X$	0.10	0.03	0.10
Indirect effect	0.25	0.28	0.25

As shown in Table 1, one-step SEM yields a proper solution but distorts the X loadings and biases the structural paths, with the direct effect of X on Y falling from 0.10 to 0.03. SAM estimates the measurement model from the indicators alone, leaving the loadings intact and recovering the true parameters.

Present Research and Contributions

This study evaluates LSAM, alongside LMS and a double-mean-centered product-indicator approach, for estimating the index of moderated mediation. It makes three contributions.

First, inference for the index has relied on the bootstrap in the limited prior work (Falk & Biesanz, 2015; Feng et al., 2020; Hayes, 2015; Tibbe & Montoya, 2022). We instead use the Monte Carlo method and evaluate how well it is calibrated.

Second, these estimators, and LSAM and LMS in particular, have not been examined for moderated mediation under structural misspecification. We evaluate them under a battery of plausible misspecifications (an omitted $W \rightarrow Y$ effect, an omitted moderated direct effect, an omitted cross-loading, and an omitted residual covariance, each at two magnitudes), the omissions that induce interpretational confounding under one-step estimation.

Third, LSAM has not been evaluated in a mediation setting or for derived quantities such as the indirect effect $a\,b$ or the index a_3b, which are nonlinear functions of the structural parameters whose behavior need not match that of their constituents. We assess all three estimators across realistic sample sizes, two reliability levels, and nonnormal latent predictors, on bias, root-mean-square error, coverage, and the Type I error and power of the index, and close with practical recommendations for estimator choice.

Simulation Design and Performance Evaluation

In this section we describe the design of the simulation study and the measures used to evaluate performance, following recent recommendations for the conduct and reporting of simulation studies (Morris et al., 2019; Pawel et al., 2025; Siepe et al., 2024).

Population Model and Estimand

We study a moderated mediation model among four latent variables: an exogenous predictor X, a moderator W, a mediator M, and an outcome Y. The structural model is as follows

$$M = a_1 X + a_2 W + a_3 (X \times W) + \zeta_M, \qquad Y = b\,M + c'\,X + \zeta_Y,$$

The primary estimands are the index of moderated mediation (imm) (Hayes, 2015), i.e., $a_3 b$. The remaining structural coefficients (a_1, a_2, b, c') and the twelve freely estimated factor loadings serve as secondary estimands. Because the index is a derived quantity rather than a free parameter, its true value is computed as $a_3 b$ implied by the correctly specified model, and each estimator's recovery of that value is assessed across conditions.

Population values were fixed at $a_1 = 0.4$, $a_2 = 0.3$, $b = 0.5$, and $c' = 0.15$, with $\mathrm{cor}(X, W) = 0.3$. The interaction coefficient a_3 is a design factor (see below). These coefficients, together with the structural disturbance variances, were chosen so that under the baseline condition (normal exogenous predictors and a correctly specified model) each dependent latent variable has $R^2 = 0.30$ at the interaction level $a_3 = 0.20$. Note that, because a_3 contributes to the variance of M, the explained variance is naturally higher at $a_3 = 0.40$ and lower at $a_3 = 0$. Each latent variable is measured by four congeneric indicators with loadings 1.0, 0.8, 0.7, and 0.6. Figure 1 depicts the population model, where the structural misspecifications studied below are drawn in grey.

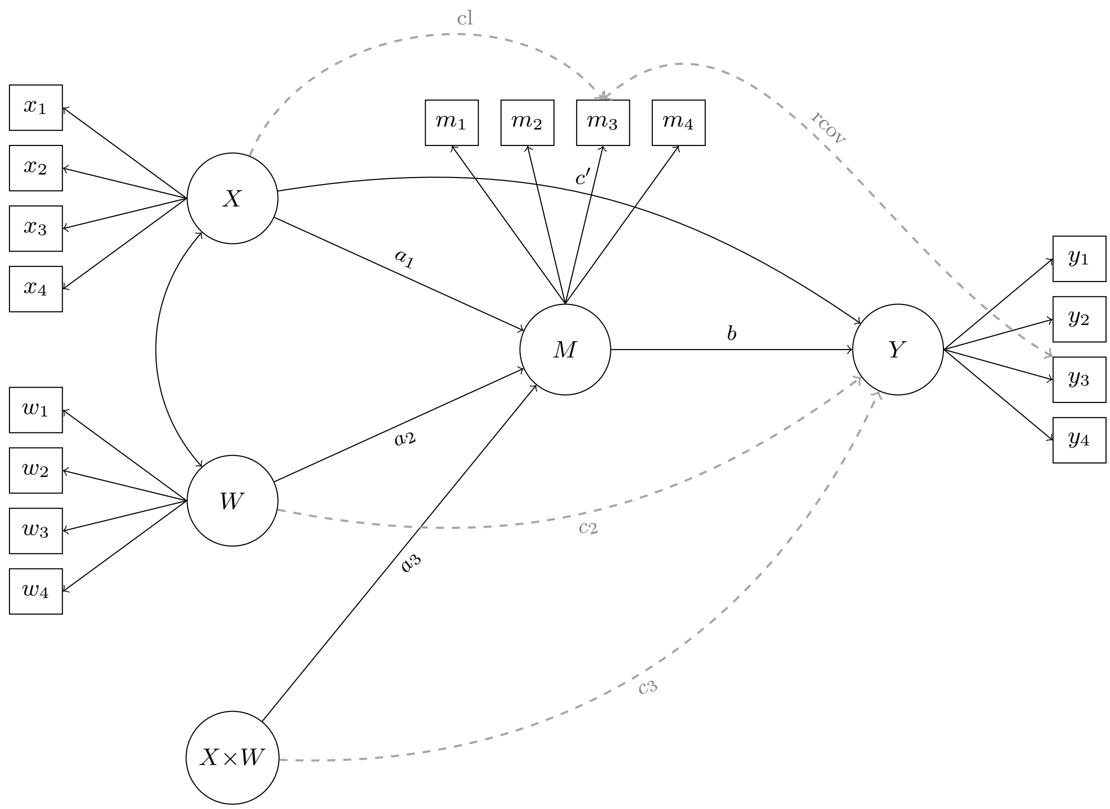
Simulation Conditions

We employed a fully factorial design crossing five factors: sample size ($n \in \{100, 200, 300, 500\}$), interaction magnitude ($a_3 \in \{0, 0.20, 0.40\}$), measurement reliability (low = 0.5, high = 0.7), the distribution of the exogenous latent predictors (normal, uniform, t, and two right-skewed χ^2 distributions), and the (mis)specification condition (nine levels; see below). This yielded $4 \times 3 \times 2 \times 5 \times 9 = 1{,}080$ conditions, each replicated 1,000 times.

Sample sizes span the small-to-moderate range typical of applied moderated mediation research, where interaction terms are estimated with limited power (Hayes, 2015; Preacher et al., 2007). The interaction magnitudes include a null condition ($a_3 = 0$) to assess Type I error and false-positive coverage of the index, and two nonzero levels representing small and moderate latent moderation effects. Reliability levels of 0.5 and 0.7 were chosen because measurement

Figure 1

Population (Data-Generating) Model for the Simulation Study.



Note. *Note*. X, W, M, and Y are latent variables, each measured by four congeneric indicators; circles are latent variables and $X{\times}W$ is the latent interaction. Black elements form the correctly specified model that every estimator fits (a_1, a_2, a_3, b, c', and the X–W covariance). Grey elements are the misspecifications present in the population but omitted from the fitted model: two in the structural part, an omitted moderator effect on the outcome (c_2) and an omitted moderated direct effect (c_3); and two in the measurement part, an omitted cross-loading of X on m_3 (cl) and an omitted residual covariance between m_3 and y_3 (rcov). Residual variances and intercepts are not shown.

error is amplified in product terms and is therefore especially consequential for latent interaction estimation (Bollen, 1989; Brandt et al., 2014), and because lower reliability is common in applied settings than the values typically examined in prior work. Nonnormality of the latent predictors is both empirically common and methodologically central for latent interaction estimators, whose distributional assumptions differ across approaches (Cham et al., 2012; Klein & Moosbrugger, 2000; Klein & Muthén, 2007). The misspecification conditions follow the moderated-mediation framework for the two structural omissions (Preacher et al., 2007) and prior latent-interaction simulation work for the two measurement omissions (Brandt et al., 2020). They were chosen as interpretable threats, acting respectively as a measured mediator–outcome confounder, a conflated moderated direct effect, exposure-correlated measurement error, and an unmeasured residual confounder (Edwards & Lambert, 2007; VanderWeele, 2015). The nine levels are detailed below.

Estimators

We compared a two-step estimator against two system-wide estimators. The two-step approach is LSAM (Rosseel et al., 2025; Rosseel & Loh, 2024). In this case, the interacting predictors X and W shared a single measurement block, while M and Y formed their own blocks. The two system-wide estimators are LMS (Klein & Moosbrugger, 2000) and UPI with double mean centering (Kelava & Brandt, 2009; Lin et al., 2010; Marsh et al., 2004).

Data Generation

Appendix A describes the complete data-generating mechanism in detail. Briefly, the latent variables X and W were drawn from a bivariate distribution with $\mathrm{Cor}(X, W) = \rho = 0.3$, constructed with the VITA (vine-to-anything) algorithm (Foldnes & Grønneberg, 2015; Grønneberg et al., 2022). Five exogenous distributions were examined, each standardized to mean 0 and variance 1 so that branches differ in shape rather than scale or location: normal, $X, W \sim \mathcal{N}(0, 1)$, using a forced Gaussian copula. The remaining four combine non-normal margins with a non-Gaussian vine copula: uniform, t_5, and two $\chi^2(2)$ conditions with X and W skewed in the same or in opposite directions. Given that, M and Y were generated from the structural equations above with independent Gaussian disturbances $\zeta_M \sim \mathcal{N}(0, \psi_M)$ and

$\zeta_Y \sim \mathcal{N}(0, \psi_Y)$, and each latent variable's four indicators were formed as $z_{\eta j} = \lambda_j \eta + \delta_{\eta j}$, with loadings $\lambda = (1, 0.8, 0.7, 0.6)$ and Gaussian errors $\delta_{\eta j} \sim \mathcal{N}(0, \theta_{\eta j})$ whose variances $\theta_{\eta j}$ were scaled to the target reliability $\omega \in \{0.5, 0.7\}$.

The nine (mis)specification conditions comprise one correctly specified condition and four families of misspecification, each at two empirically calibrated magnitudes (standardized 0.25 and 0.50, detailed in Appendix A). Two are structural: an omitted direct effect of the moderator on the outcome (`c2`) and an omitted moderated direct effect (`c3`). The others are in the measurement model: a cross-loading of a mediator indicator on X (`cl`) and a residual covariance between a mediator and an outcome indicator (`rcov`). In every case the effect is present in the population but omitted from the fitted model.

Nonconvergence, Outliers, and Inadmissible Solutions

Following Lonati et al. (2024), replications were screened for estimation failures and inadmissible solutions before analysis. Because the compared estimators are implemented across different packages, we applied an identical admissibility check to all of them: a solution was flagged when any residual (unique) variance was nonpositive (a Heywood case) or when the latent covariance matrix was not positive definite. Each estimator was fit independently within a replication, so a failure of one method produced a missing row for that method only, with the captured error or warning retained for diagnostics.

Performance Measures

Performance was assessed with the measures implemented in the *simhelpers* package (Joshi & Pustejovsky, 2025), following recommendations in Morris et al. (2019). For the index of moderated mediation and the structural coefficients we report absolute and relative bias, RMSE, the mean standard error and the SE/SD ratio, confidence-interval coverage, and—where applicable—the rejection rate (Type I error when $a_3 = 0$, power otherwise). Monte Carlo uncertainty in each performance measure is quantified with jackknife Monte Carlo standard errors (Joshi & Pustejovsky, 2025).

Inference for the index of moderated mediation accounts for the skewness of the product

$a_3 b$: for $r = 1, \ldots, R$ (with $R = 20{,}000$) we drew

$$\left(a_3^{(r)}, b^{(r)}\right) \sim \mathcal{N}_2\!\left((\hat{a}_3, \hat{b}),\, \hat{\Sigma}_{a_3 b}\right), \qquad \mathrm{imm}^{(r)} = a_3^{(r)}\, b^{(r)},$$

where $\hat{\Sigma}_{a_3 b}$ is the estimated covariance of $(\hat{a}_3, \hat{b})$, and took the 2.5th and 97.5th empirical percentiles of $\{\mathrm{imm}^{(r)}\}_{r=1}^{R}$ as the interval. The structural coefficients and loadings use Wald (?) intervals from the model-implied parameter covariance matrix. The same implementation was applied to all three estimators for comparability.

Reproducibility and Computational Details

All analyses were conducted in R (R Core Team, 2025) using the *lavaan* (Rosseel, 2012) and *modsem* (Slupphaug et al., 2024) packages for estimation, *covsim* (Grønneberg et al., 2022) and *rvinecopulib* (Nagler & Vatter, 2025) for data generation, and *simhelpers* (Joshi & Pustejovsky, 2025) for performance evaluation. Conditions were run in parallel with R's *parallel* package; reproducible and independent random-number streams across workers were obtained with the L'Ecuyer-CMRG generator seeded from a single master seed. The computational environment was pinned with Nix via the *rix* package (Rodrigues & Baumann, 2026) to ensure reproducibility of the computational environment, and the article was written with the *apaquarto* extension (Schneider, 2026). All code and materials are available at the project repository.

References

Algina, J., & Moulder, B. C. (2001). A note on estimating the Jöreskog–Yang model for latent variable interaction using LISREL 8.3. *Structural Equation Modeling: A Multidisciplinary Journal*, *8*(1), 40–52. https://doi.org/10.1207/S15328007SEM0801_3

Arminger, G., & Muthén, B. O. (1998). A Bayesian approach to nonlinear latent variable models using the Gibbs sampler and the Metropolis–Hastings algorithm. *Psychometrika*, *63*(3), 271–300. https://doi.org/10.1007/bf02294856

Asparouhov, T., & Muthén, B. (2021). Bayesian estimation of single and multilevel models with latent variable interactions. *Structural Equation Modeling: A Multidisciplinary Journal*, *28*(2), 314–328. https://doi.org/10.1080/10705511.2020.1761808

Bohrnstedt, G. W., & Marwell, G. (1978). The reliability of products of two random variables. *Sociological Methodology*, *9*, 254–273. https://doi.org/10.2307/270812

Boker, S. M., Oertzen, T. von, Pritikin, J. N., Hunter, M. D., Brick, T. R., Brandmaier, A. M., & Neale, M. C. (2023). Products of variables in structural equation models. *Structural Equation Modeling: A Multidisciplinary Journal*, *30*(5), 708–718. https://doi.org/10.1080/10705511.2022.2141749

Bollen, K. A. (1989). *Structural equations with latent variables*. John Wiley & Sons. https://doi.org/10.1002/9781118619179

Bollen, K. A. (1995). Structural equation models that are nonlinear in latent variables: A least-squares estimator. *Sociological Methodology*, *25*, 223–251. https://doi.org/10.2307/271068

Bollen, K. A., & Hoyle, R. H. (2012). Latent variables in structural equation modeling. In R. H. Hoyle (Ed.), *Handbook of structural equation modeling* (pp. 56–67). The Guilford Press.

Brandt, H., Kelava, A., & Klein, A. G. (2014). A simulation study comparing recent approaches for the estimation of nonlinear effects in SEM under the condition of nonnormality. *Structural Equation Modeling: A Multidisciplinary Journal*, *21*(2), 181–195. https://doi.org/10.1080/10705511.2014.882660

Brandt, H., Umbach, N., Kelava, A., & Bollen, K. A. (2020). Comparing estimators for latent interaction models under structural and distributional misspecifications. *Psychological Methods*, *25*(3), 321–345. https://doi.org/10.1037/met0000231

Burt, R. S. (1976). Interpretational confounding of unobserved variables in structural equation models. *Sociological Methods & Research*, *5*(1), 3–52. https://doi.org/10.1177/004912417600500101

Busemeyer, J. R., & Jones, L. E. (1983). Analysis of multiplicative combination rules when the causal variables are measured with error. *Psychological Bulletin*, *93*(3), 549–562. https://doi.org/10.1037/0033-2909.93.3.549

Cham, H., West, S. G., Ma, Y., & Aiken, L. S. (2012). Estimating latent variable interactions with nonnormal observed data: A comparison of four approaches. *Multivariate Behavioral Research*, *47*(6), 840–876. https://doi.org/10.1080/00273171.2012.732901

Charlton, A., Montoya, A. K., Price, J., Hilgard, J., & Krefeld-Schwalb, A. (2021). *Noise in the process: An assessment of the evidential value of mediation effects in marketing journals*. PsyArXiv preprint. https://doi.org/10.31234/osf.io/ck2r5

Cole, D. A., & Preacher, K. J. (2014). Manifest variable path analysis: Potentially serious and misleading consequences due to uncorrected measurement error. *Psychological Methods*, *19*(2), 300–315. https://doi.org/10.1037/a0033805

Cox, K., & Kelcey, B. (2021). Croon's bias-corrected estimation of latent interactions. *Structural Equation Modeling: A Multidisciplinary Journal*, *28*(6), 863–874. https://doi.org/10.1080/10705511.2021.1922283

Edwards, J. R., & Lambert, L. S. (2007). Methods for integrating moderation and mediation: A general analytical framework using moderated path analysis. *Psychological Methods*, *12*(1), 1–22. https://doi.org/10.1037/1082-989X.12.1.1

Falk, C. F., & Biesanz, J. C. (2015). Inference and interval estimation methods for indirect effects with latent variable models. *Structural Equation Modeling: A Multidisciplinary Journal*, *22*(1), 24–38. https://doi.org/10.1080/10705511.2014.935266

Feng, Q., Song, Q., Zhang, L., Zheng, S., & Pan, J. (2020). Integration of moderation and mediation in a latent variable framework: A comparison of estimation approaches for the second-stage moderated mediation model. *Frontiers in Psychology*, *11*, 2167. https://doi.org/10.3389/fpsyg.2020.02167

Foldnes, N., & Grønneberg, S. (2015). How general is the Vale–Maurelli simulation approach? *Psychometrika*, *80*(4), 1066–1083. https://doi.org/10.1007/s11336-014-9414-0

Gonzalez, O., & MacKinnon, D. P. (2021). The measurement of the mediator and its influence on statistical mediation conclusions. *Psychological Methods*, *26*(1), 1–17. https://doi.org/10.1037/met0000263

Grønneberg, S., Foldnes, N., & Marcoulides, K. M. (2022). covsim: An R package for simulating non-normal data for structural equation models using copulas. *Journal of Statistical Software*, *102*(3), 1–45. https://doi.org/10.18637/jss.v102.i03

Hayes, A. F. (2015). An index and test of linear moderated mediation. *Multivariate Behavioral Research*, *50*(1), 1–22. https://doi.org/10.1080/00273171.2014.962683

Hayes, A. F. (2018). *Introduction to mediation, moderation, and conditional process analysis: A regression-based approach* (2nd ed.). The Guilford Press.

Holst, K. K., & Budtz-Jørgensen, E. (2020). A two-stage estimation procedure for non-linear structural equation models. *Biostatistics*, *21*(4), 676–691. https://doi.org/10.1093/biostatistics/kxy082

Hoyle, R. H., & Kenny, D. A. (1999). Sample size, reliability, and tests of statistical mediation. In R. H. Hoyle (Ed.), *Statistical strategies for small sample research* (pp. 195–222). Sage.

Imai, K., Keele, L., & Tingley, D. (2010). A general approach to causal mediation analysis. *Psychological Methods*, *15*(4), 309–334. https://doi.org/10.1037/a0020761

Jöreskog, K. G., & Yan, F. (1996). Nonlinear structural equation models: The Kenny–Judd model with interaction effects. In G. A. Marcoulides & R. E. Schumacker (Eds.), *Advanced structural equation modeling* (pp. 57–88). Lawrence Erlbaum Associates.

Joshi, M., & Pustejovsky, J. (2025). *simhelpers: Helper functions for simulation studies*. https://CRAN.R-project.org/package=simhelpers

Kelava, A., & Brandt, H. (2009). Estimation of nonlinear latent structural equation models using the extended unconstrained approach. *Review of Psychology*, *16*(2), 123–131.

Keller, B. T. (2025). A factored regression approach to modeling latent variable interactions and nonlinear effects. *Psychological Methods*. https://doi.org/10.1037/met0000777

Kenny, D. A., & Judd, C. M. (1984). Estimating the nonlinear and interactive effects of latent variables. *Psychological Bulletin*, *96*(1), 201–210. https://doi.org/10.1037/0033-2909.96.1.201

Klein, A. G., & Moosbrugger, H. (2000). Maximum likelihood estimation of latent interaction effects with the LMS method. *Psychometrika*, *65*(4), 457–474.

https://doi.org/10.1007/BF02296338

Klein, A. G., & Muthén, B. O. (2007). Quasi-maximum likelihood estimation of structural equation models with multiple interaction and quadratic effects. *Multivariate Behavioral Research*, *42*(4), 647–673.

Ledgerwood, A., & Shrout, P. E. (2011). The trade-off between accuracy and precision in latent variable models of mediation processes. *Journal of Personality and Social Psychology*, *101*(6), 1174–1188. https://doi.org/10.1037/a0024776

Lin, G.-C., Wen, Z., Marsh, H. W., & Lin, H.-S. (2010). Structural equation models of latent interactions: Clarification of orthogonalizing and double-mean-centering strategies. *Structural Equation Modeling: A Multidisciplinary Journal*, *17*(3), 374–391. https://doi.org/10.1080/10705511.2010.488999

Lonati, S., Rönkkö, M., & Antonakis, J. (2024). Normality assumption in latent interaction models. *Psychological Methods*. https://doi.org/10.1037/met0000657

Marsh, H. W., Wen, Z., & Hau, K.-T. (2004). Structural equation models of latent interactions: Evaluation of alternative estimation strategies and indicator construction. *Psychological Methods*, *9*(3), 275–300. https://doi.org/10.1037/1082-989X.9.3.275

Montoya, A. K. (2024). Combining statistical and causal mediation analysis. In H. T. Reis, T. West, & C. M. Judd (Eds.), *Handbook of research methods in social and personality psychology* (pp. 622–652). Cambridge University Press.

Mooijaart, A., & Bentler, P. M. (2010). An alternative approach for nonlinear latent variable models. *Structural Equation Modeling: A Multidisciplinary Journal*, *17*(3), 357–373. https://doi.org/10.1080/10705511.2010.488997

Mooijaart, A., & Satorra, A. (2012). Moment testing for interaction terms in structural equation modeling. *Psychometrika*, *77*(1), 65–84. https://doi.org/10.1007/s11336-011-9232-6

Morris, T. P., White, I. R., & Crowther, M. J. (2019). Using simulation studies to evaluate statistical methods. *Statistics in Medicine*, *38*(11), 2074–2102. https://doi.org/10.1002/sim.8086

Nagler, T., & Vatter, T. (2025). *rvinecopulib: High performance algorithms for vine copula modeling*. https://CRAN.R-project.org/package=rvinecopulib

Ng, J. C. K., & Chan, W. (2020). Latent moderation analysis: A factor score approach. *Structural Equation Modeling: A Multidisciplinary Journal*, *27*(4), 629–648. https://doi.org/10.1080/10705511.2019.1664304

Pawel, S., Bartoš, F., Siepe, B. S., & Lohmann, A. (2025). Handling missingness, failures, and non-convergence in simulation studies: A review of current practices and recommendations. *The American Statistician*, 1–18. https://doi.org/10.1080/00031305.2025.2540002

Pieters, R. (2017). Meaningful mediation analysis: Plausible causal inference and informative communication. *Journal of Consumer Research*, *44*(3), 692–716. https://doi.org/10.1093/jcr/ucx081

Preacher, K. J., Rucker, D. D., & Hayes, A. F. (2007). Addressing moderated mediation hypotheses: Theory, methods, and prescriptions. *Multivariate Behavioral Research*, *42*(1), 185–227. https://doi.org/10.1080/00273170701341316

R Core Team. (2025). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing. https://www.R-project.org/

Rodrigues, B., & Baumann, P. (2026). *Rix: Reproducible data science environments with Nix*. https://CRAN.R-project.org/package=rix

Rohrer, J. M., Hünermund, P., Arslan, R. C., & Elson, M. (2022). That's a lot to process! Pitfalls of popular path models. *Advances in Methods and Practices in Psychological Science*, *5*(2), 1–14. https://doi.org/10.1177/25152459221095827

Rosseel, Y. (2012). lavaan: An R package for structural equation modeling. *Journal of Statistical Software*, *48*(2), 1–36. https://doi.org/10.18637/jss.v048.i02

Rosseel, Y., Burghgraeve, E., Loh, W. W., & Schermelleh-Engel, K. (2025). Structural after measurement (SAM) approaches for accommodating latent quadratic and interaction effects. *Behavior Research Methods*, *57*(4), 101. https://doi.org/10.3758/s13428-024-02532-y

Rosseel, Y., & Loh, W. W. (2024). A structural after measurement approach to structural equation modeling. *Psychological Methods*, *29*(3), 561–588. https://doi.org/10.1037/met0000503

Schneider, W. J. (2026). *apaquarto: A Quarto extension for creating APA style (7th edition) documents*. https://github.com/wjschne/apaquarto

Siepe, B. S., Bartoš, F., Morris, T. P., Boulesteix, A.-L., Heck, D. W., & Pawel, S. (2024). Simulation studies for methodological research in psychology: A standardized template for planning, preregistration, and reporting. *Psychological Methods*. https://doi.org/10.1037/met0000695

Slupphaug, K. S., Mehmetoglu, M., & Mittner, M. (2024). modsem: An R package for estimating latent interactions and quadratic effects. *Structural Equation Modeling: A Multidisciplinary Journal*. https://doi.org/10.1080/10705511.2024.2417409

Sterner, P., Pargent, F., Deffner, D., & Goretzko, D. (2024). A causal framework for the comparability of latent variables. *Structural Equation Modeling: A Multidisciplinary Journal*, *31*(5), 747–758. https://doi.org/10.1080/10705511.2024.2339396

Tibbe, T. D., & Montoya, A. K. (2022). Correcting the bias correction for the bootstrap confidence interval in mediation analysis. *Frontiers in Psychology*, *13*, 810258. https://doi.org/10.3389/fpsyg.2022.810258

VanderWeele, T. J. (2015). *Explanation in causal inference: Methods for mediation and interaction*. Oxford University Press.

VanderWeele, T. J., & Vansteelandt, S. (2022). A statistical test to reject the structural interpretation of a latent factor model. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, *84*(5), 2032–2054. https://doi.org/10.1111/rssb.12555

Wall, M. M., & Amemiya, Y. (2000). Estimation for polynomial structural equation models. *Journal of the American Statistical Association*, *95*(451), 929–940. https://doi.org/10.2307/2669475

Wall, M. M., & Amemiya, Y. (2001). Generalized appended product indicator procedure for nonlinear structural equation analysis. *Journal of Educational and Behavioral Statistics*, *26*(1), 1–29. https://doi.org/10.3102/10769986026001001

Wall, M. M., & Amemiya, Y. (2003). A method of moments technique for fitting interaction effects in structural equation models. *British Journal of Mathematical and Statistical Psychology*, *56*(1), 47–63. https://doi.org/10.1348/000711003321645331